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CIT 371: Linux System Administration

Lab 01: Software Installation

Introduction:

In this lab, you will learn how to install software on Linux. You will first use a package manager, then you will install software manually.

Part 0: Mandatory VM Snapshots

Instructions:

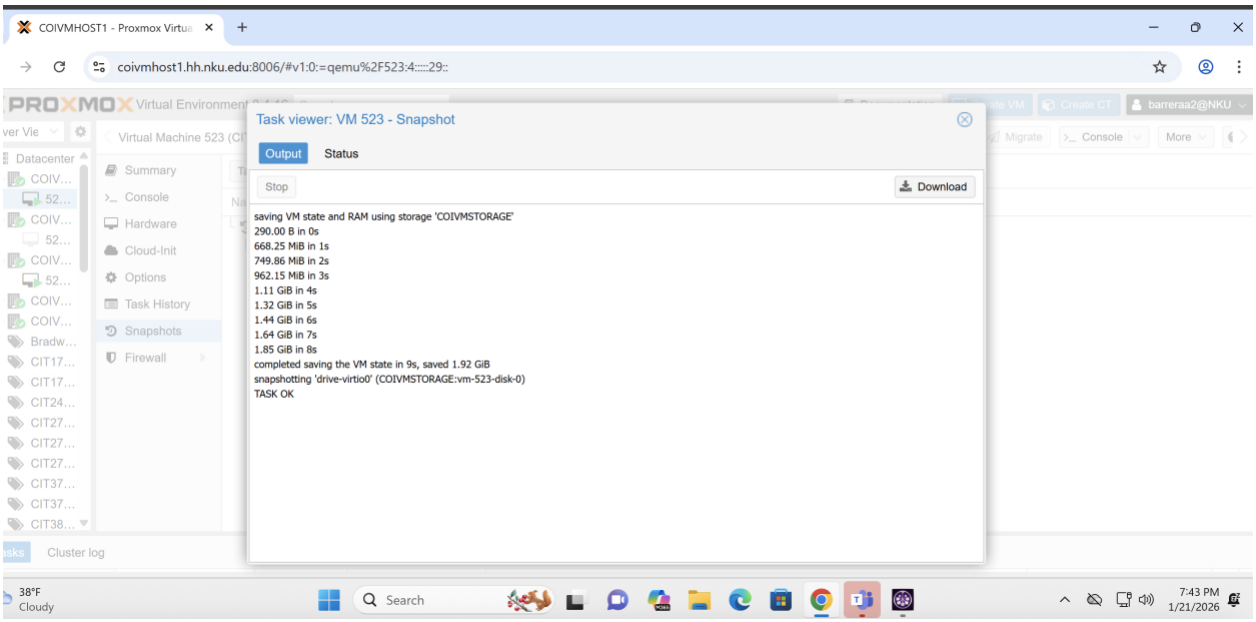
Before you begin this lab, **<u>if no Day 1 snapshot is present</u>**, you are required to take a snapshot of the following VM(s) (for this lab, you can skip the snapshot if a Day 1 one is present):

- **Ubuntu Server**

<u>Failure to create this snapshot and provide proof of it will result in a 15-point reduction on your lab score</u>. This steep penalty is to prevent this class from burdening our virtual environment managers with VM replacement requests. Snapshots will save them and you time by allowing you to revert to a known good state if something breaks in your VMs.

Screenshot Requirements:

Take a screenshot of the snapshot(s) of the VM(s) you took (or the Day 1 snapshot if present).



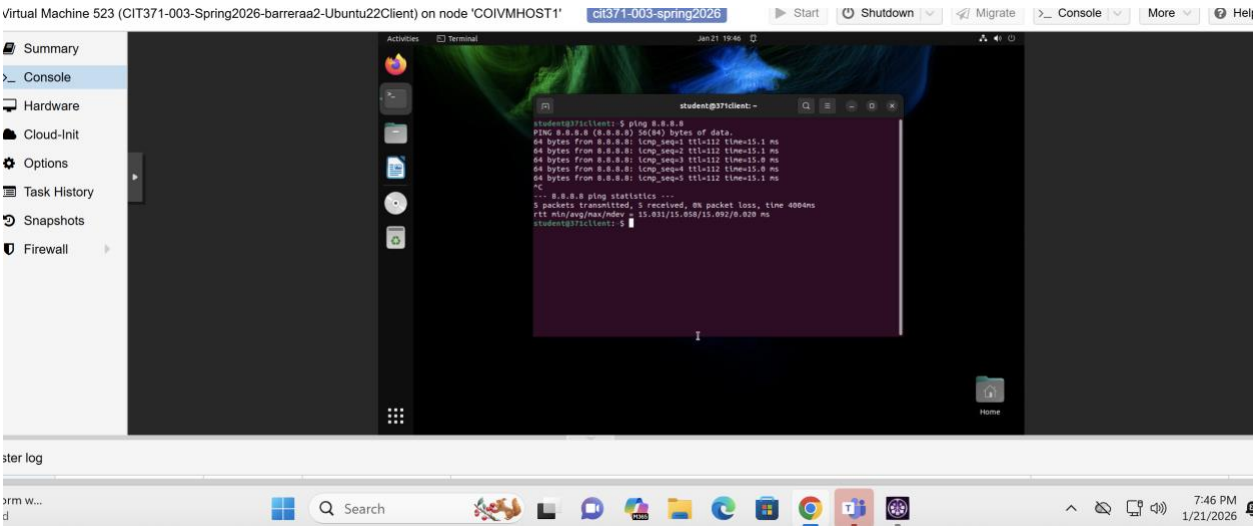
Part 1: Preparing Your Lab Environment

Instructions:

To start this lab, sign into the COI Proxmox cluster, then turn on your PfSense VM. **Wait a couple minutes, then turn on your <u>Ubuntu server</u>.**

After your Ubuntu server boots, sign in as the student user with *Student371!* as the password. Open the terminal and run a command that tests whether you can reach the address 8.8.8.8, which is Google's public DNS server. Press Ctrl + C after the command sends out about 4 or 5 transmissions.

If you can successfully reach the address, your VM has Internet. If the command does not reach it, troubleshoot by making sure your PfSense router is on.



Part 2: Package Manager Method

Instructions:

On your Ubuntu server, install the tree program using the built-in command line package manager for Ubuntu. You will need to research how to do this. As a hint, the program's installation can be performed with a single command.

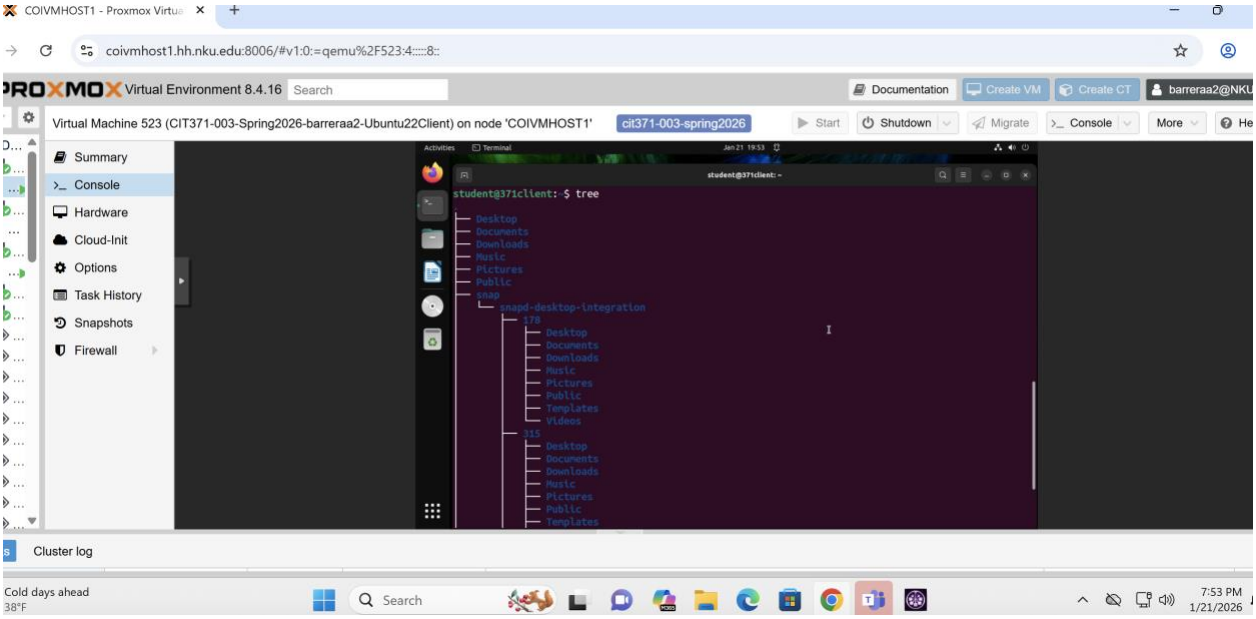
After installing it, run the *tree* command in /home/student.

> **NOTE:** While attempting to install the program when you first turn on your VM, you may receive the following error:

> If you see this error, please wait a few minutes before proceeding. Your Ubuntu server is installing updates in the background and will need to finish before you continue.

Screenshot Requirements:

Take one or more screenshots of the output of the *tree* command. Consider decreasing the terminal's zoom percentage to capture the output in just one screenshot.



Part 3: Manual Method

On your Ubuntu server, use the Option 2 section of [this article from Cherry Servers]() to manually install Git **from source** (the dependencies will be installed with a package manager; Git itself will not be). After installing Git from source, configure it with your name and NKU email.

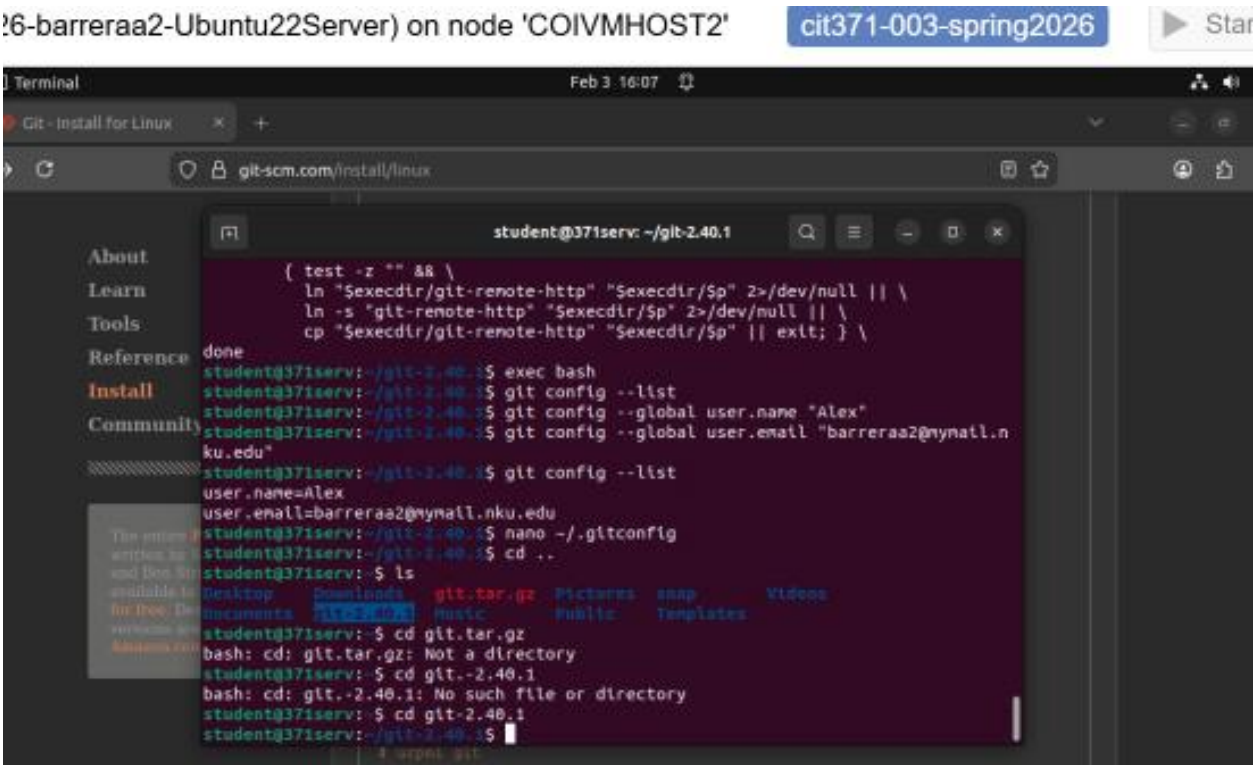
The following are some troubleshooting tips if you run into issues:

- If you receive error messages that start with, "Failed to fetch…" while attempting to install the dependencies, run the following command to run updates on the server:
    - *sudo apt update*
- On **Step 4 – Navigate to the Extracted Directory** in the article, if the *cd git-** command does not work, run the *ls* command to verify the Git directory is present. If it is, you can Tab complete the directory name by typing *cd git* and then pressing the Tab key.

```
student@371serv:~$ ls
Desktop  Documents  Downloads  git-2.40.1  git.tar.gz  Music  Pictures  Public  snap  Templates  Videos
student@371serv:~$
student@371serv:~$ cd git-2.40.1/
student@371serv:~/git-2.40.1$
```

Screenshot Requirements:

Take a screenshot of the *git config –list* command being run after completing the article, <u>showing your name and NKU email</u>.



Submission Instructions:

No wiki article is required for this lab. Instead, just submit a Word document containing the following:

- Your name and NKU email.
- The screenshots requested in this document's Screenshots Requirements sections.